

- 1 (a) Underline **all** the vector quantities in the following list:

distance

displacement

speed

velocity

acceleration.

[1]

- (b) A car of mass 1400 kg is initially travelling horizontally in a straight line at a constant speed of 15 ms^{-1} . The brakes are applied and the car decelerates uniformly to rest in a distance of 35 m.

Calculate:

- (i) the initial kinetic energy of the car

kinetic energy = J [2]

- (ii) the acceleration of the car

acceleration = ms^{-2} [2]

- (iii) the resultant force on the decelerating car.

force = N [2]



- (c) The initial speed of the car in **1(b)** is now doubled to 30 ms^{-1} . The brakes of the car are then applied and the car comes to rest. The resultant force on the car remains the same as in **1(b)(iii)**.

Describe and explain the effect, if any, of doubling the initial speed on:

- (i) the magnitude of the acceleration of the car

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..... [1]

- (ii) the distance travelled by the car in coming to rest.

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..... [2]

[Total: 10]